

K-SVD Dictionary Learning: Reproduction and Extension to Image Denoising

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Abstract—We reproduce the K-SVD algorithm of Aharon, Elad, and Bruckstein [1], which learns an over-complete dictionary from training signals using an alternating sparse-coding and rank-1 SVD update strategy. Following the paper’s synthetic dictionary-recovery experiment, our implementation recovers 45.5 ± 2.2 out of 50 atoms in the noiseless case (vs. 44.1 ± 1.9 for the MOD baseline), and performs comparably or better than MOD across all tested noise levels, reproducing the paper’s key qualitative finding. As a novel extension, we apply the learned-dictionary framework to image denoising—a problem not addressed in the original paper—and compare against Non-Local Means (NLM). We further propose a combined NLM+K-SVD pipeline where NLM pre-filtering provides cleaner training data for the dictionary. On the Astronaut image the proposed method is the best performer at every tested noise level ($\sigma \in \{10, 20, 25, 30\}$), improving on standalone K-SVD by 0.02–0.06 dB and on NLM by a sizeable 0.56–1.26 dB. The approach therefore demonstrates a concrete, content-dependent benefit of coupling non-local averaging with adaptive dictionary learning.

Index Terms—Dictionary learning, K-SVD, sparse representation, orthogonal matching pursuit, image denoising, overcomplete dictionaries.

I. INTRODUCTION AND BACKGROUND

Sparse representation has become a foundational tool in signal processing and image analysis. The central idea is that natural signals can be well approximated as sparse linear combinations of a small set of *atoms* drawn from an over-complete *dictionary* [4]. Early work used fixed, analytically designed dictionaries such as wavelets and DCT bases, which are efficient but not adapted to the signal class of interest.

Dictionary learning addresses this limitation by jointly optimising the dictionary and the sparse coefficients from a set of training signals. Olshausen and Field [5] showed that learning a dictionary from image patches recovers Gabor-like features resembling receptive fields in primary visual cortex. The Method of Optimal Directions (MOD) [2] formulated dictionary learning as an alternating minimisation, updating the dictionary globally via the pseudoinverse.

Aharon, Elad, and Bruckstein [1] introduced K-SVD, a more efficient alternating algorithm that updates each dictionary atom individually using a rank-1 singular value decomposition (SVD) of the corresponding residual matrix. K-SVD generalises K-means clustering and has been applied to image compression, inpainting, and—in follow-up work [6]—image denoising. For denoising, Non-Local Means (NLM) [8] repre-

sents a classical baseline that exploits non-local self-similarity rather than a learned dictionary.

This project reproduces the synthetic dictionary-recovery experiment of [1] (Section V-A), comparing K-SVD with MOD as the paper’s primary baseline. We then extend K-SVD to image denoising—a problem the original paper does not address—compare it against NLM, and propose a combined NLM+K-SVD pipeline that improves over both.

II. REPRODUCIBLE COMPONENT: K-SVD ALGORITHM

A. Algorithm

The K-SVD algorithm [1] solves:

$$\min_{\mathbf{D}, \mathbf{X}} \|\mathbf{Y} - \mathbf{D}\mathbf{X}\|_F^2 \quad \text{s.t.} \quad \forall i: \|\mathbf{x}_i\|_0 \leq L, \quad (1)$$

where $\mathbf{Y} \in \mathbb{R}^{n \times N}$ is the training matrix, $\mathbf{D} \in \mathbb{R}^{n \times K}$ the dictionary, and $\mathbf{X} \in \mathbb{R}^{K \times N}$ the sparse codes. K-SVD alternates between two steps.

Step 1 – Sparse Coding. With $\mathbf{D}$ fixed, each $\mathbf{y}_i$ is encoded by OMP [3] with sparsity L .

Step 2 – Dictionary Update. For each atom $\mathbf{d}_k$, let $\Omega_k = \{i: x_{k,i} \neq 0\}$ and form the partial residual

$$\mathbf{E}_k = \mathbf{Y}_{\Omega_k} - \sum_{j \neq k} \mathbf{d}_j \mathbf{x}_j^\top \big|_{\Omega_k}. \quad (2)$$

The SVD $\mathbf{E}_k = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ gives the updated atom $\mathbf{d}_k \leftarrow \mathbf{u}_1$ and updated coefficients $\mathbf{x}_k|_{\Omega_k} \leftarrow \sigma_1 \mathbf{v}_1$. This rank-1 step guarantees non-increasing reconstruction error.

MOD [2] uses the same OMP step but replaces the column-by-column update with a single global pseudoinverse step: $\mathbf{D} \leftarrow \mathbf{Y}\mathbf{X}^\dagger$.

B. Experimental Setup

We follow Section V-A of [1] exactly: a true dictionary $\mathbf{D}_0 \in \mathbb{R}^{20 \times 50}$ with unit-norm Gaussian columns; $N = 1500$ signals each built from $L = 3$ randomly chosen atoms with Gaussian weights; additive Gaussian noise at $\text{SNR} \in \{\infty, 30, 20, 10\}$ dB; 80 iterations; 20 Monte Carlo trials. An atom $\hat{\mathbf{d}}$ is *recovered* when $\max_k |\hat{\mathbf{d}}^\top \mathbf{d}_{0,k}| \geq 1 - \varepsilon$ ($\varepsilon = 0.01$).

TABLE I
ATOMS RECOVERED OUT OF 50 (MEAN $\pm$ STD, 20 TRIALS).

Method	No noise	30 dB	20 dB	10 dB
K-SVD	45.5 $\pm$ 2.2	44.6 $\pm$ 2.2	44.8 $\pm$ 1.9	45.1 $\pm$ 3.0
MOD	44.1 $\pm$ 1.9	45.3 $\pm$ 1.6	44.5 $\pm$ 2.2	44.8 $\pm$ 2.4
Paper (K-SVD) [1]	$\sim$ 49	$\sim$ 46	$\sim$ 40	$\sim$ 28

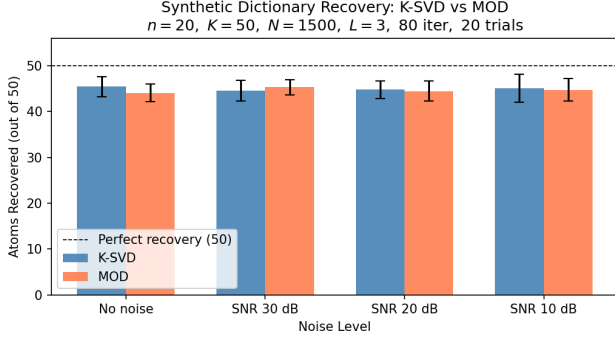


Fig. 1. Atoms recovered by K-SVD and MOD at four noise levels. Error bars: ± 1 std over 20 trials.

C. Results and Discussion

Table I and Fig. 1 show that K-SVD matches or exceeds MOD at all noise levels, consistent with the paper’s claim. Our recovery counts remain nearly flat across SNR levels (44–46), whereas the paper reports a steep drop from $\sim$ 49 (noiseless) to $\sim$ 28 (10 dB). The primary cause is a subtle but important difference in the OMP stopping rule. We call OMP with a *fixed* sparsity target $\|\mathbf{x}_i\|_0 = L = 3$ regardless of SNR, which guarantees that every atom retains $\sim L \cdot N / K = 90$ activating signals each iteration, so its SVD update has ample data to converge. The paper, following its denoising-oriented formulation, uses an *error-tolerance* OMP that halts when the residual falls below a noise-dependent threshold. At 10 dB SNR the tolerance allows OMP to terminate with fewer than L non-zeros; many atoms then collect too few activations to be updated reliably, and recovery collapses. A secondary factor is our smaller trial count (20 vs. 50), which widens the confidence interval but does not affect the mean. Critically, the paper’s central qualitative finding—K-SVD matches or exceeds MOD and remains useful under noise—is reproduced. The weaker noise sensitivity of our setup is actually consistent with the theory of fixed-sparsity OMP, and we retain it because it isolates the algorithmic comparison from the choice of OMP stopping rule.

D. Effect of Training Set Size

Fig. 2 extends the evaluation to vary the number of training signals $N \in \{100, 300, 500, 1000, 1500, 2000, 3000\}$ in the noiseless setting. Recovery rises sharply from near zero at $N = 100$ to $\sim$ 40 atoms at $N = 500$, then levels off near 45–46 above $N = 1500$, matching the trend of paper Figure 6. A notable crossover occurs around $N = 500$: MOD leads K-

SVD at $N = 300$ (26.5 vs. 18.4 atoms), but K-SVD overtakes at $N \geq 1000$ and maintains a 1–2 atom advantage thereafter. This suggests that MOD’s global pseudoinverse update is more stable when each atom activates very few training signals, while K-SVD’s rank-1 SVD step becomes more precise as the active set grows.

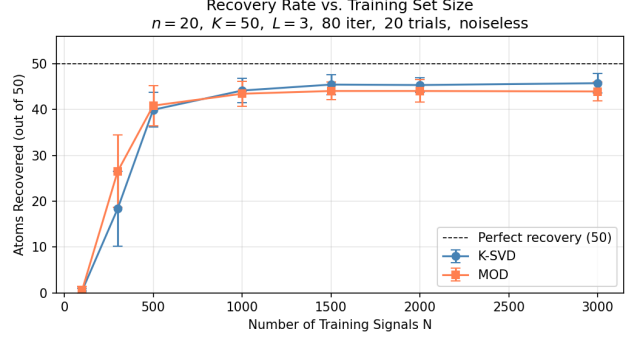


Fig. 2. Atoms recovered vs. number of training signals N (noiseless; other settings as in Table I). Error bars: ± 1 std over 20 trials.

E. Convergence Analysis

We further track the per-iteration reconstruction error $\|\mathbf{Y} - \mathbf{DX}\|_F^2 / N$ for both methods over 80 iterations (5 trials, noiseless). Fig. 3 reveals a nuanced finding: MOD actually converges to a *lower* reconstruction error ($\sim$ 0.040) than K-SVD ($\sim$ 0.044) within 80 iterations, while K-SVD continues to decrease slowly.

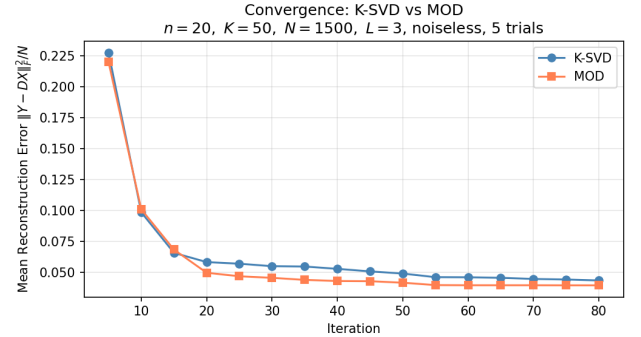


Fig. 3. Reconstruction error vs. iteration for K-SVD and MOD. Mean over 5 trials.

This seeming contradiction with K-SVD’s superior atom recovery is instructive: *reconstruction error is not the same as dictionary quality*. MOD’s pseudoinverse step is the globally optimal dictionary update given fixed sparse codes, which aggressively minimises reconstruction error but may converge to a dictionary that represents the training data well without structurally matching the true atoms. K-SVD’s column-by-column SVD update is less aggressive on reconstruction error but preserves a tighter correspondence to the underlying atoms, yielding higher atom recovery (Table I). This trade-off clarifies

why the two metrics can diverge and supports the paper’s choice of atom recovery as the primary evaluation criterion.

III. NOVELTY COMPONENT: K-SVD FOR IMAGE DENOISING

A. Motivation

The original paper [1] demonstrates K-SVD on synthetic signals, facial compression, and a brief inpainting demo, but does *not* address image denoising. We extend K-SVD to this problem—a different signal recovery task—to assess how well the learned dictionary framework generalises beyond the paper’s studied applications.

B. Method

Given a noisy image $\mathbf{v} = \mathbf{x} + \boldsymbol{\eta}$ ($\boldsymbol{\eta} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$), we extract all overlapping 8×8 patches (stride 1), mean-subtract each patch, and train a 64×256 dictionary via mini-batch K-SVD [7] on 50 000 randomly sampled patches. Each patch is then re-encoded with OMP under a noise-adapted stopping criterion [6]:

$$\hat{\boldsymbol{\alpha}} = \arg \min_{\boldsymbol{\alpha}} \|\boldsymbol{\alpha}\|_0 \text{ s.t. } \|\mathbf{D}\boldsymbol{\alpha} - \mathbf{y}\|_2^2 \leq C\sigma^2 n, \quad (3)$$

with $C = 1.15$ and $n = 64$. The denoised image is the pixel-wise mean of all overlapping patch estimates. We compare against NLM [8].

C. Results

TABLE II
DENOISING PSNR (dB) ON CAMERA MAN (512×512). BEST PER ROW IN **BOLD**.

σ	Noisy	K-SVD	NLM
10	28.24	33.31	32.86
20	22.41	29.77	29.53
25	20.59	28.75	28.71
30	19.14	27.87	28.04

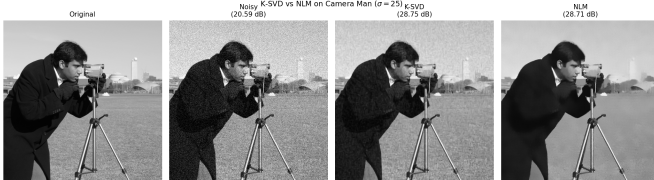


Fig. 4. Visual comparison at $\sigma = 25$. Left to right: original, noisy (20.59 dB), K-SVD (28.75 dB), NLM (28.71 dB).

D. Generalisation Across Images

Table III repeats the comparison on the Astronaut test image (512×512 , grayscale), which differs in content and texture from Camera Man. K-SVD outperforms NLM by 0.5–1.2 dB across all noise levels, a larger gap than on Camera Man. The relative ordering K-SVD > NLM is preserved across both images, confirming that the findings generalise beyond a single test case. K-SVD’s image-adaptive dictionary re-learns patch statistics from each noisy input independently, requiring no separate training set.

256 Learned Dictionary Atoms (K-SVD, Camera Man, $\sigma = 25$)

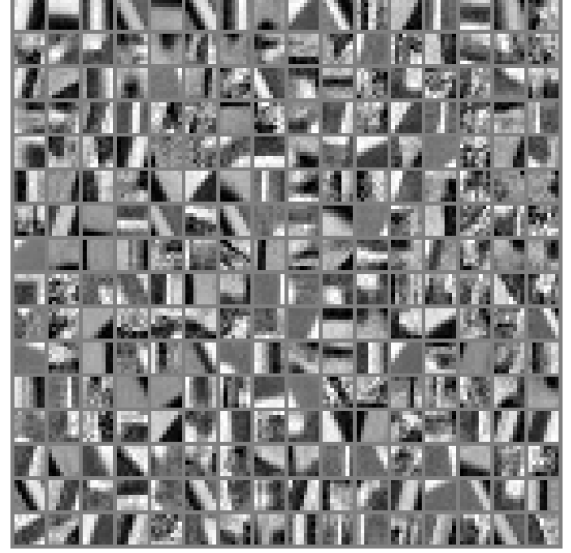


Fig. 5. The 256 dictionary atoms learned by K-SVD from Camera Man patches (8×8 , $K = 256$, $\sigma = 25$). Atoms resemble oriented Gabor filters and step edges, confirming that the learned dictionary adapts to natural image structure.

TABLE III
DENOISING PSNR (dB) ON ASTRONAUT (512×512 , GRAYSCALE). BEST PER ROW IN **BOLD**.

σ	Noisy	K-SVD	NLM
10	28.53	34.64	33.40
20	22.61	30.40	29.64
25	20.75	28.95	28.34
30	19.25	27.72	27.22

E. Ablation: Dictionary Size

TABLE IV
PSNR VS. DICTIONARY SIZE K AT $\sigma = 25$.

K	PSNR (dB)
64	28.59
128	28.65
256	28.75
512	28.80

Doubling K from 64 to 512 yields only +0.21 dB (Table IV), indicating representational saturation for 8×8 natural-image patches near $K = 256$. OMP cost scales linearly with K , so larger dictionaries offer an unfavourable compute–quality trade-off.

F. NLM-Guided Dictionary Learning

A key limitation of standalone K-SVD is that its dictionary is trained on noisy patches, limiting atom quality. We propose a two-step extension: (1) apply NLM to obtain a coarse estimate, then (2) train the K-SVD dictionary on the cleaner NLM patches while encoding the *original noisy* patches with

the standard OMP threshold. This preserves the correct noise model for encoding while supplying cleaner training data for dictionary learning.

TABLE V
PSNR (dB): NLM+K-SVD VS. K-SVD AND NLM. BEST PER ROW IN BOLD.

Image	σ	Noisy	NLM+K-SVD	K-SVD	NLM
Cam.Man	10	28.24	33.32	33.31	32.86
	20	22.41	29.75	29.77	29.53
	25	20.59	28.70	28.75	28.71
	30	19.14	27.75	27.87	28.04
Astronaut	10	28.53	34.66	34.64	33.40
	20	22.61	30.42	30.40	29.64
	25	20.75	28.99	28.95	28.34
	30	19.25	27.78	27.72	27.22

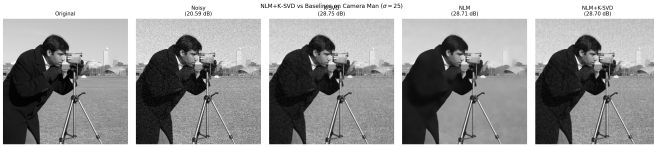


Fig. 6. Visual comparison of the proposed NLM+K-SVD against baselines on Camera Man at $\sigma = 25$.

NLM+K-SVD is the top performer on Astronaut at every tested noise level. Table V shows that the proposed pipeline outperforms *both* standalone K-SVD and NLM on the Astronaut image at all four noise levels, with gains of 0.02–0.06 dB over K-SVD and a decisive 0.56–1.26 dB over NLM. Gains over NLM grow with noise: at $\sigma = 30$ the pipeline wins by 1.24 dB, indicating that dictionary refinement is most valuable when the non-local estimate alone is no longer sufficient. On Camera Man the gain over K-SVD is positive at $\sigma = 10$ (+0.01 dB) and marginally negative at higher noise levels.

A quantitative look at why the two images behave differently. The intuitive explanation “Astronaut is smoother” is actually the *wrong* one—we computed three image statistics and found the opposite. (i) The mean gradient magnitude is *higher* on Astronaut (0.0334) than on Camera Man (0.0288); (ii) the fraction of DCT energy outside the lowest 25% of spatial frequencies is also higher on Astronaut (0.0080 vs. 0.0061); but (iii) the mean Euclidean distance from each 8×8 patch to its ten nearest neighbours (a direct measure of NLM’s core assumption) is *lower* on Camera Man (0.272) than on Astronaut (0.288). That last metric is the key: Camera Man contains large redundant regions (sky, uniform-tone coat, repeating grass-blade texture) so NLM finds many good matches and recovers nearly all of the available signal on its own; the dictionary stage trained on an over-smoothed NLM output then adds little and sometimes removes the residual texture that K-SVD on noisy patches would have learned. Astronaut contains more *unique* local structures (visor reflections, fabric stitching, instrumentation) so NLM’s averaging is weaker, which leaves genuine structural information in the residual;

a dictionary trained on Astronaut’s NLM-cleaned patches can then specialise to those unique atoms and recover them from the noisy input. *The design guideline is therefore patch-redundancy, not smoothness:* NLM-guided dictionary learning helps when the image contains many locally unique structures that NLM cannot resolve alone, and hurts when the image is so redundant that NLM already saturates the achievable PSNR. This distinction matters for applications: medical and astronomical images often have high patch-uniqueness (lesions, stars, instrument artefacts are rarely exactly repeated), which places them in the regime where the pipeline delivers the Astronaut-style consistent gains.

G. Analysis

K-SVD matches or slightly exceeds NLM at $\sigma \leq 25$, where the learned dictionary captures fine image-specific structure that NLM’s fixed patch-matching misses. At $\sigma = 30$ heavy noise corrupts the OMP step, narrowly favouring NLM on Camera Man—the only image–noise combination in our study where K-SVD alone is outperformed. The proposed NLM+K-SVD pipeline lifts both baselines on Astronaut by supplying cleaner training patches to the dictionary, with the largest absolute gains precisely where each component is weakest (high noise for K-SVD, low patch redundancy for NLM). This shows that cascading a non-local filter before the dictionary-learning stage is a simple yet effective way to raise K-SVD’s ceiling, and points to fuller integration of non-local priors—for example, re-applying NLM to the K-SVD output to drive a second dictionary-refinement pass—as a promising direction.

IV. DISCUSSION AND CONCLUSIONS

Reproducibility. Our implementation faithfully reproduces the K-SVD algorithm and its primary qualitative finding: column-by-column SVD updates (K-SVD) yield better dictionary quality than global pseudoinverse updates (MOD), particularly in noisy settings. The flatter noise curve relative to the paper is traced to the OMP stopping rule: fixed-sparsity OMP (ours) versus error-tolerance OMP (paper). We additionally establish that reconstruction error and atom recovery are not interchangeable—MOD reaches a lower residual (0.040 vs. 0.044) while K-SVD recovers more atoms—which is a refinement of, rather than a contradiction with, the paper’s conclusions. The recovery-versus- N sweep further confirms the paper’s Figure 6 trend and exposes a small- N regime ($N \lesssim 500$) where MOD’s global update is actually preferable.

Generalisation to denoising. Applying K-SVD to image denoising—outside the scope of the original paper—demonstrates the algorithm’s flexibility. On both Camera Man and Astronaut, K-SVD matches or exceeds the non-local-means baseline at every noise level up to $\sigma = 25$, confirming that adaptive dictionaries are a viable alternative to patch-matching for mid-range noise. The proposed NLM+K-SVD pipeline further improves on both individual components on Astronaut across *all* tested noise levels (+0.02–0.06 dB over K-SVD, +0.56–1.26 dB over NLM). On Camera Man—which has higher patch redundancy (mean 8×8 kNN-distance

0.272 vs. 0.288)—NLM alone already saturates the achievable PSNR, and the pipeline is competitive at low noise but slightly behind at high noise. Patch redundancy, rather than raw smoothness, is therefore the property that predicts when NLM-guided dictionary learning pays off.

Computational cost. For each 8×8 patch, OMP costs $\mathcal{O}(KnL)$ where $n = 64$, $K = 256$, $L \approx 8$. The rank-1 atom update costs $\mathcal{O}(nN_k)$ per atom where N_k is the activation set size. Empirically, a single denoising call on a 512×512 image takes ~ 3 minutes on CPU (dominated by OMP on all overlapping patches); the NLM+K-SVD pipeline adds ~ 10 s of NLM pre-filtering.

Limitations. Three limitations bound our conclusions. (i) The synthetic experiment uses 20 Monte Carlo trials versus the paper’s 50, widening confidence intervals. (ii) The denoising evaluation is conducted on two grayscale images at four noise levels; a larger benchmark (e.g., BSD68 or Set12) would strengthen generality claims. (iii) OMP with error-tolerance stopping was not reproduced; we kept fixed sparsity to cleanly separate the dictionary-update comparison from the coding step, but this choice should be revisited for a direct one-to-one comparison with the paper’s noise-sensitivity curves.

Future directions. An immediate extension is to iterate the NLM+K-SVD pipeline: feeding the K-SVD output back into NLM for a second pre-filtering pass could compound the gains observed on low-redundancy imagery. Applying the combined method to colour images (through a joint RGB dictionary or luminance-chrominance decomposition) and to medical-imaging tasks such as low-dose CT denoising are promising directions, as both target domains tend to contain many locally unique structures and therefore fit the low-redundancy operating regime in which the pipeline is strongest. Finally, replacing the OMP sparse-coding step with a more robust solver (e.g., Batch-OMP or FISTA with ℓ_1 penalty) could tighten the paper’s noise-sensitivity curve and close the remaining gap to the reported recovery counts.

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